

Analyzing a clip for activity 4

The small clip I've chosen is clipped from segment 2, with timestamp from 8:06:54 to 8:06:58. The clip is showing the running animation, then he grabs a ledge and hop over it.

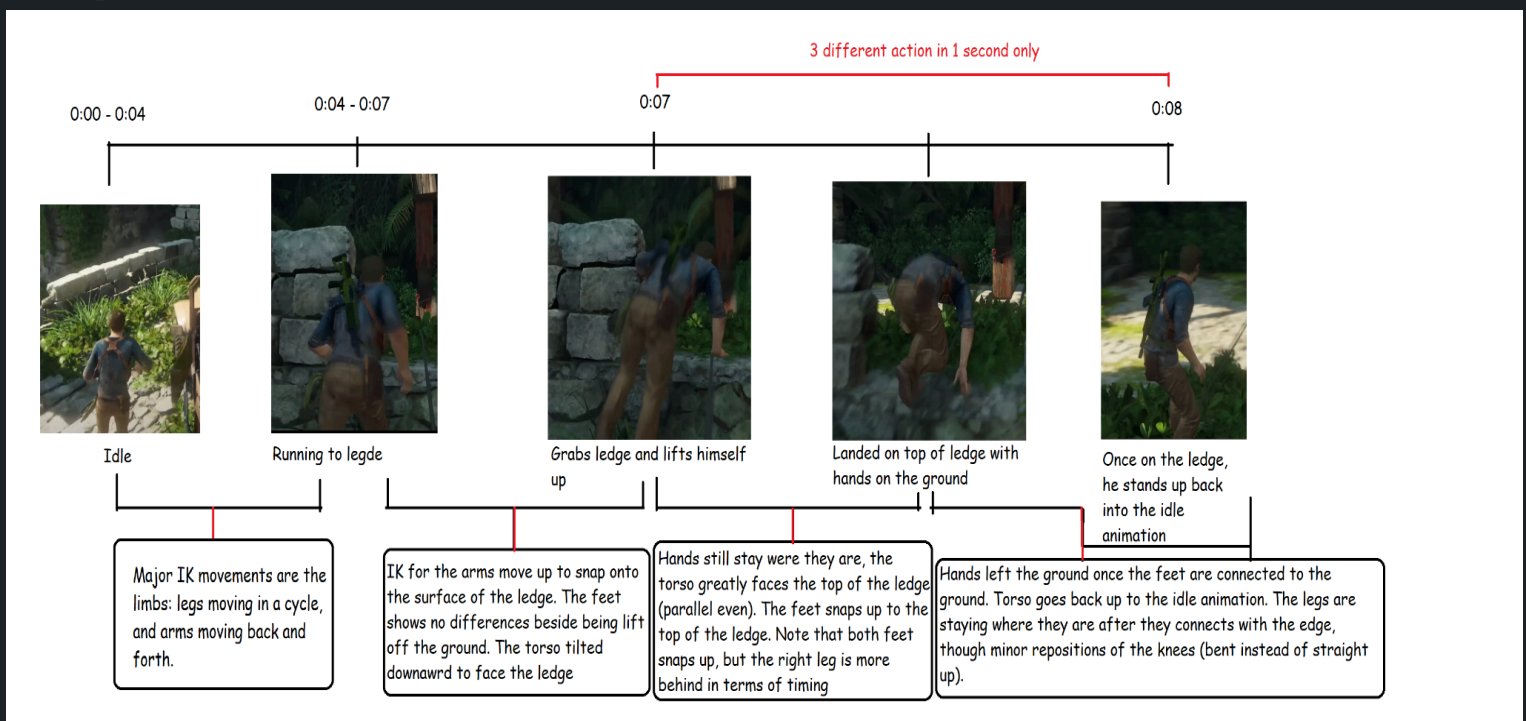
From that segment, there were several key IK usage, which involves with his limbs the most: legs for running, and arms for grabbing the ledge as well as pulling him over it. There is also the head and overall torso moving to look and face the ledge as well.

Potential targets are definitely the hands and feet, and minor targets such as the head when looking around as well as the torso (chest target) when turning.

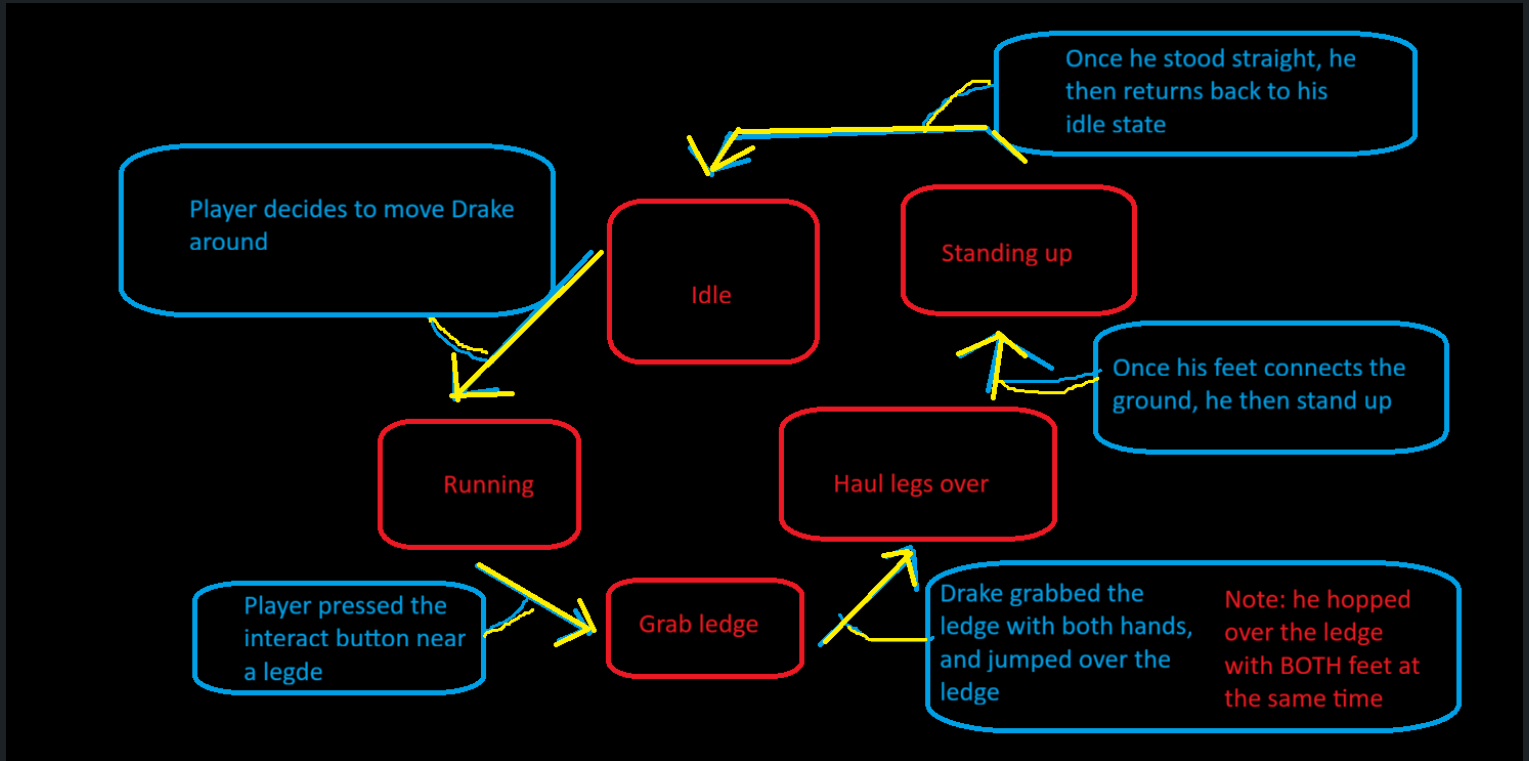
There would be 4 poles for the elbows and knees (2 upper arms, 2 forearm bones, 2 thighs, 2 shins, and two feet bones), as well as 2 more for the fingers (all 28 finger bones). 1 pole for the spine (3 bones: head, chest, and hip total).

The poles are for the joint to face towards when bending (ie: the elbow would usually go towards the backside when the hand is bent closer to the torso, so the pole for that elbow would need to be behind the character).

Timing chart:



State diagram:



Very simple and linear movements for the states.

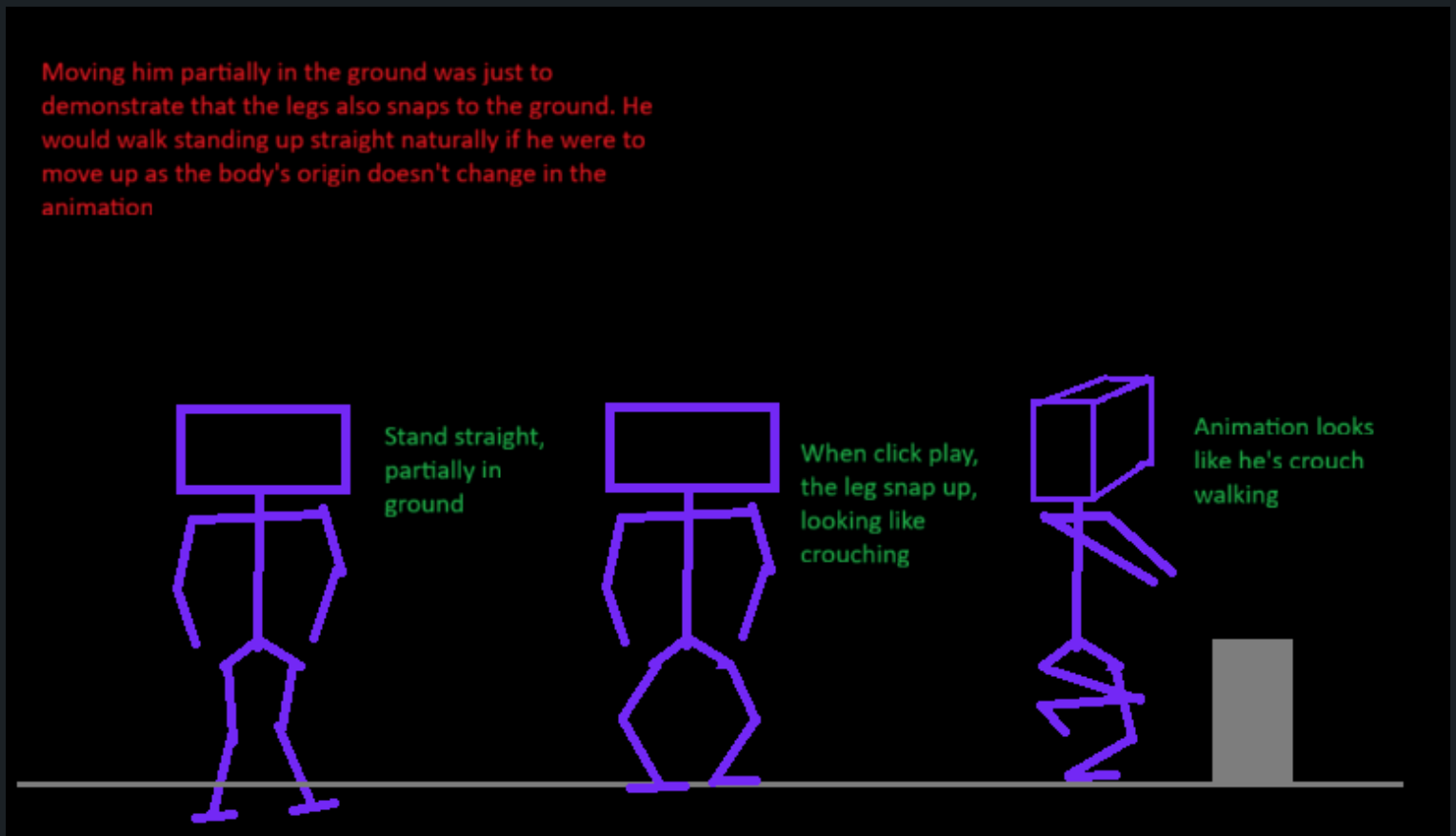
Two constraints that I will be animating are the legs for the walking animation and the arms that move to the ledge to get up on a block.

For the development side of things, I studied and referenced the simple IK animation script included in the package to see how the character can be moved, adding on more features such as animating both hands so that it both points at the barrier and rotation axis for leg walking movement.

The resulting script is basically just transforming the target's position (in this case the vertical y axis) of each limb so that it will match the object with colliders (in this case the barrier and the ground).

For the character, I had microwaveman positioned under the floor at the start to demonstrate that the legs would snap onto the ground surface (it looks like he is crouch-walking with the bent knees). Using Raycast, I cast rays from the feet so that when it touches the object with the collider, something will happen, in this case I keep the feet upward to avoid the barrier. For the overall running movement, I just parented the feet target to two spinning cylinders to

simulate up and down movement of the legs, this would keep the legs constantly moving but also help demonstrate the feet collision detection better.



I'd rate this 10/10 for extreme beat down to my brain, and also the sad truth aftermath of not getting it perfect (I'm still glad the raycast works though).

I guess the most difficult part was definitely positioning the targets as well as the poles so that the limbs moved naturally and realistically instead of being a jello monster with the limbs flailing all over the place. There were also the parenting issues, in which I had a really hard time correcting the pivot points, the rotation axis, as well as the overall structure in the hierarchy for the IK "skeleton."

The least challenging part was definitely adding the ik script onto the target and moving the bones around. That beginning part of the activity was actually a bit fun... (until the other part hits).

From a rewatch of the video reference, compared to my animation, well... mine's definitely different in many ways. One major thing is the overall stiffness of the animation (but that is because I didn't choose to animate the whole body as one of my two chosen constraints).

The hands, while reaching out to touch the barrier as it was coming towards him, clips through each other (which I mistakenly did not account for: I should've been using two different targets instead of one), as well as rotated at an odd angle (I'm guessing is another parenting issue).

The head and body does move and rotate to look at the barrier, however, I believe I didn't set the head target far enough. This results in the character trying to touch his forehead to the barrier instead of looking at it from a short distance, away from his face instead of the top of his head.

The legs seem to be working fine... that is until the barrier hits. While the legs did go over the barrier, it came out of sync every time he "hops" over the barrier. This results in the character's inconsistent jumping movement: sometimes bunny hopping, and sometimes skipping over the barrier (which I believe is related to parenting issues with the rotating cylinders...).

Though there is one minor thing similar to the ref is the legs nearly line up to jump over the ledge, but even that is inconsistent as said previously. I guess the hand reaching out towards the barrier is an achievement. As well as the character at least looking at where he is grabbing.

I was a bit sad that despite trying very hard, I couldn't make it animate how I envisioned it in my mind (it was a lot cooler in my head!). But it was a bit magical to see the feet automatically move up and over the barrier, it kind of even made me feel uncanny (I'm not so sure but maybe that's because I'm used to seeing movement animated by hand instead of procedurally animated?). So while it came off really janky and weird, it still feels nice in a way since it makes the character feel more "aware" and slightly more realistic.

The most useful skill I learned while doing the activity is definitely implementing IK to a character. Even though it looks complicated, it's actually pretty simple to implement IK, since it is just transforming the bones most of the time. I haven't tried it, but I believe I can enhance the already existing walking animation for the character by making the movement smarter and realistic.

What I'd do differently next time is give the target two points instead for the hands to grab onto the barrier (the hands are spread out and not converge at a single point). For the feet, I'd try to fix the parenting issue, maybe constraint the feet to only move up and down in world space (I think I can use `space.world` for that).

I believe I can apply this type of animation to the characters for the GDW game project. This may allow them to do complex movements such as walking over bumpy ground, wall climbing on rocky areas, perform attacking animation more accurately (ex: creatures with long necks can move their head more freely and attack the player by lining their head and biting). Since this also gave me some uncanny feelings, this may even enhance the enemies' scare factor as well!